

数电实验-2.8 同步计数器应用

1. 同步计数器的基本特性分析 (8进制同步计数器)

功能表
EN=0 时, 计数器不工作.

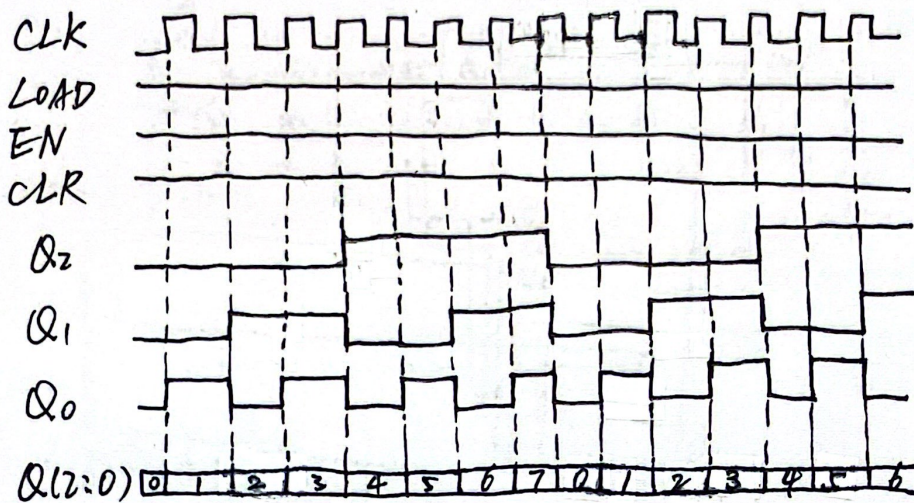
时钟信号 CLK (EN=1)	置数控制端 EN	清除控制端
(计数器为上沿触发)	LOAD=0	LOAD=1
0 → 1 上升沿	置数	跳变至下一计数状态
0 / 1 低/高电平	保持	保持
1 → 0 下降沿	保持	保持

当 LOAD=1,

而且 EN=1, CLR=1 时, 同步计数器处于正常计数状态.

波形图如下:

体现不了优先级

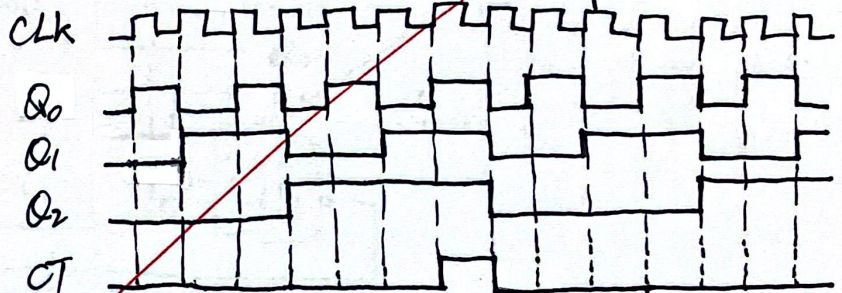
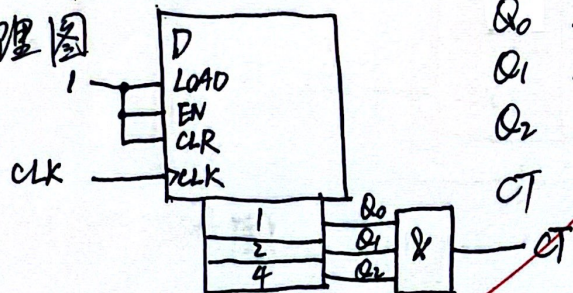


2. 构成秒信号发生器

将 EN=1, LOAD=1, CLR=1 置为高电平.

设置 CT 为进位信号.

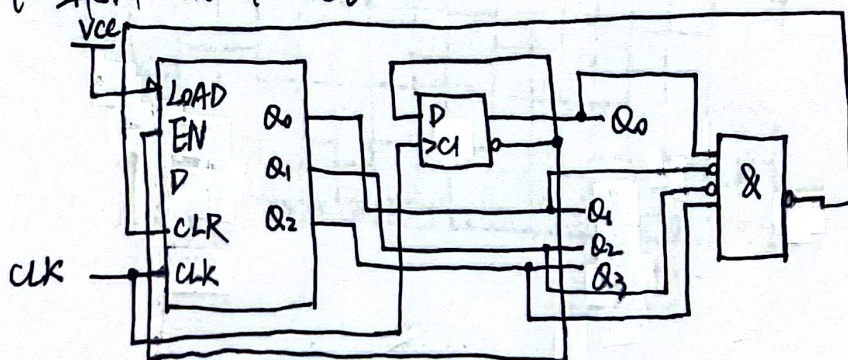
电原理图



为秒信号脉冲信号.

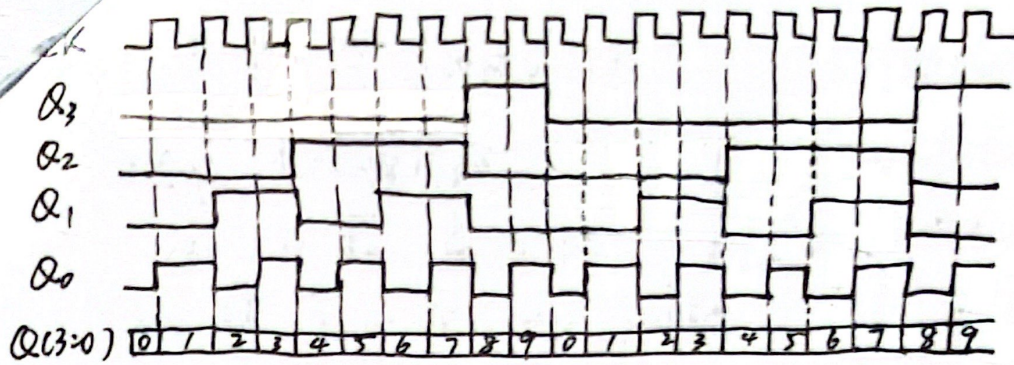
3. 10进制计数器和6进制计数器的设计

(1) 10进制加法计数器

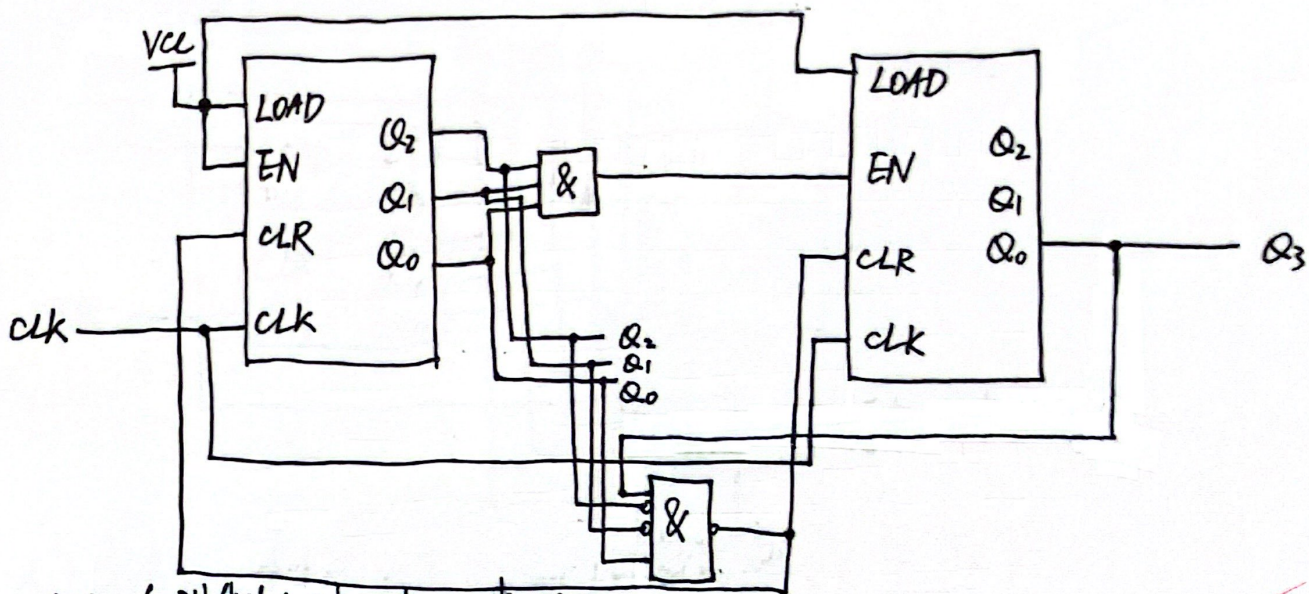


方法1: 8进制计数器 + 1个触发器作为最低位计数位

波形图 (LOAD 控制始终置为1)

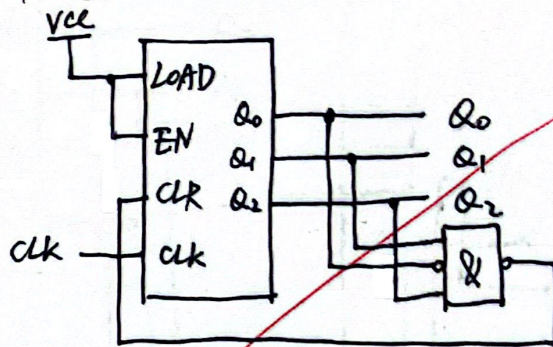


方法2: 用2个8进制计数器构成

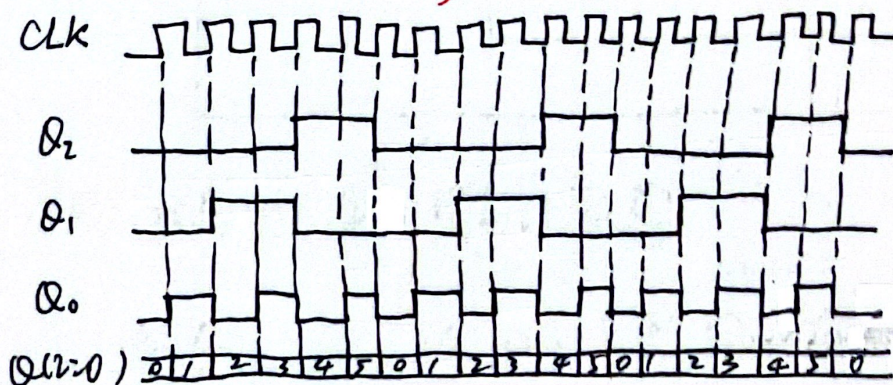


(2) 6进制加法计数器

电原理图



功能验证波形图

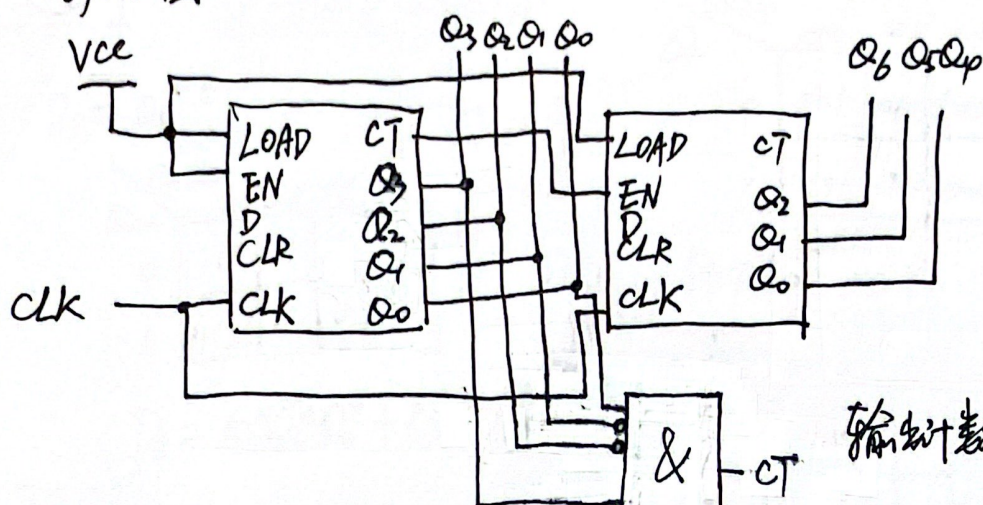


方法1. 利用6进制和10进制计数器通过拆8计数构成6进制计数器

缺点: 计数的具体数值是不连续的, 在后面仿真波形图中就能看出, 出现了0~80的不连续的计数值, 除第一个循环外, 其余均为从80开始到73结束的一个60个计数情况的循环, 实现6进制计数。

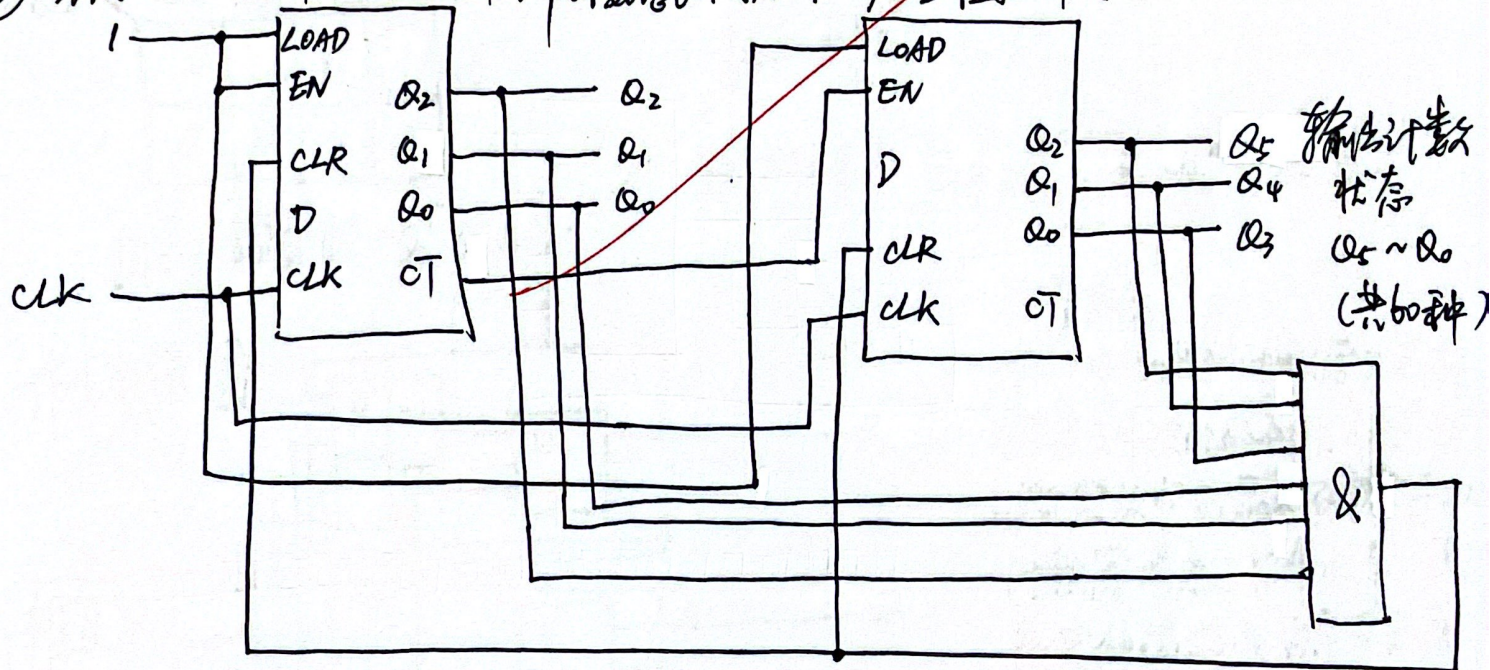
优点: 结构简单, 易于集成。

电原理图



输出计数状态为 $Q_6 \sim Q_0$ [共60种]。不考虑冗余

② 方法2. 利用2个8进制计数器构成, 电原理图如下:



输出计数状态 $Q_5 \sim Q_0$ (共60种)

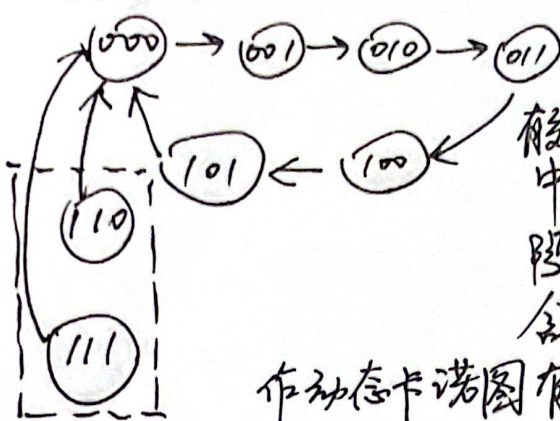
3. 带冗余状态的6进制同步计数器

$2^3 - 6 = 2$, 选用3个D触发器, 因此存在了2个冗余状态
下面利用摩尔模型进行状态机和其电路的设计。

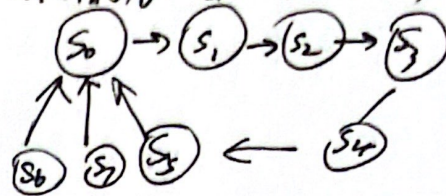
进行状态分配

根据三项原则，现态与次态的对应的关系均是唯一的，故分配为：(原则1失效)

$Q_2 Q_1 Q_0$



$Q_2 Q_1 Q_0$	00	01	11	10
0	S_0	S_1	S_3	S_2
1	S_4	S_5	S_7	S_6



有效状态中不存在不同现态指向同一次态的情况，故有效的6个状态可随意分配，但2个冗余状态的分配要到一个卡诺图的质蕴合中，因为110, 111 2个冗余态均指向000同一次态。

作动态卡诺图有：

$$D_2 = Q_2^{n+1}$$

$Q_2 Q_1 Q_0$	00	01	11	10
0	0	0	1	0
1	1	0	0	0

$$D_2 = Q_2^n \bar{Q}_1^n \bar{Q}_0^n + \bar{Q}_2^n Q_1^n Q_0^n$$

$$D_1 = Q_1^{n+1}$$

$Q_2 Q_1 Q_0$	00	01	11	10
0	0	1	0	1
1	0	0	0	0

$$D_1 = \bar{Q}_2^n \bar{Q}_1^n Q_0^n + Q_2^n Q_1^n \bar{Q}_0^n$$

$Q_2 Q_1 Q_0$	00	01	11	10
0	001	010	100	011
1	101	000	000	000

$$D_0 = Q_0^{n+1}$$

$Q_2 Q_1 Q_0$	00	01	11	10
0	1	0	0	1
1	1	0	0	0

$$D_0 = \bar{Q}_1^n \bar{Q}_0^n + Q_1^n \bar{Q}_0^n$$

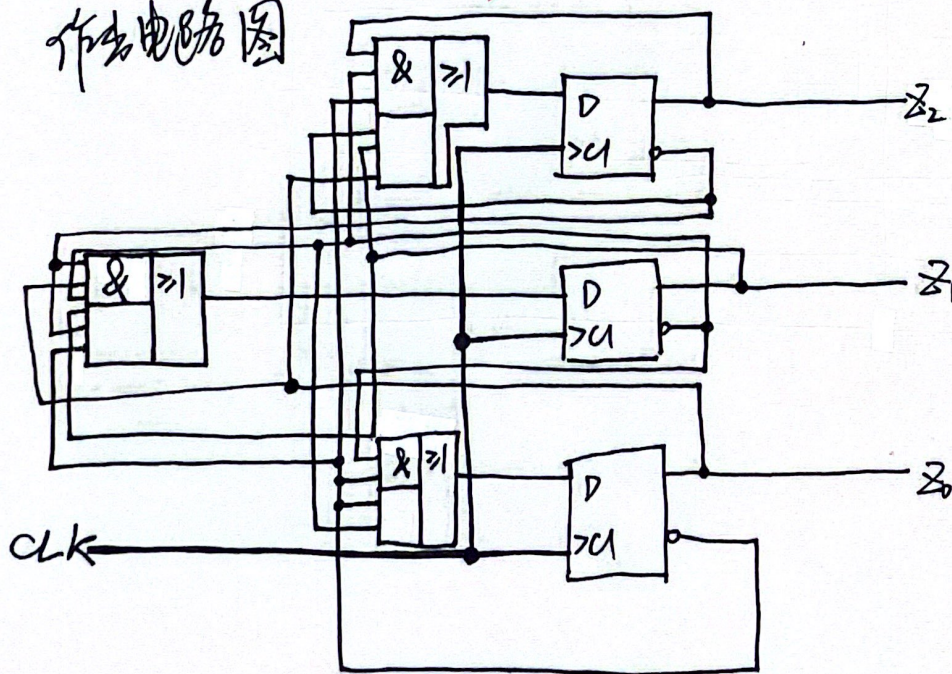
$$\text{输出方程: } Z_2 = Q_2^n, Z_1 = Q_1^n, Z_0 = Q_0^n$$

D触发器

D	$Q_n \rightarrow Q_{n+1}$
1	$d \rightarrow 1 (1)$
0	$d \rightarrow 0 (0)$

次态方程

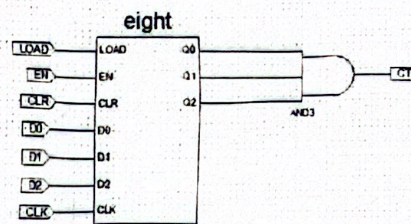
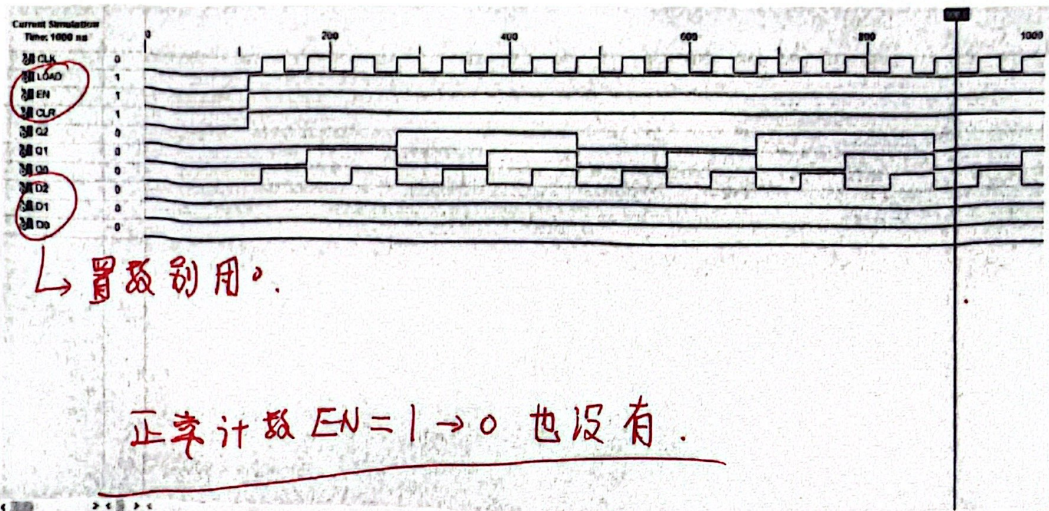
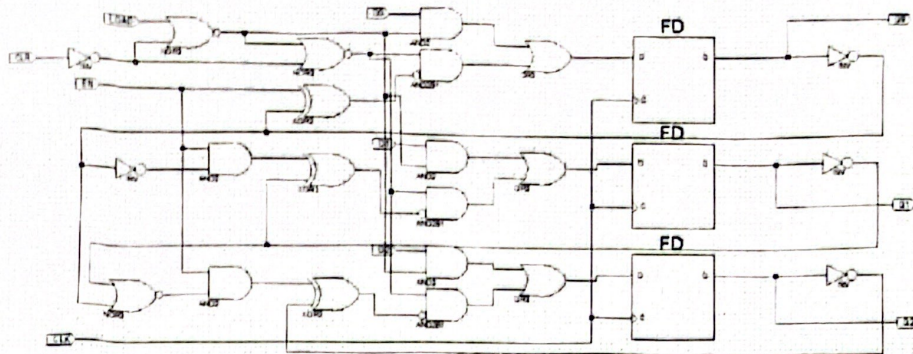
作电路图



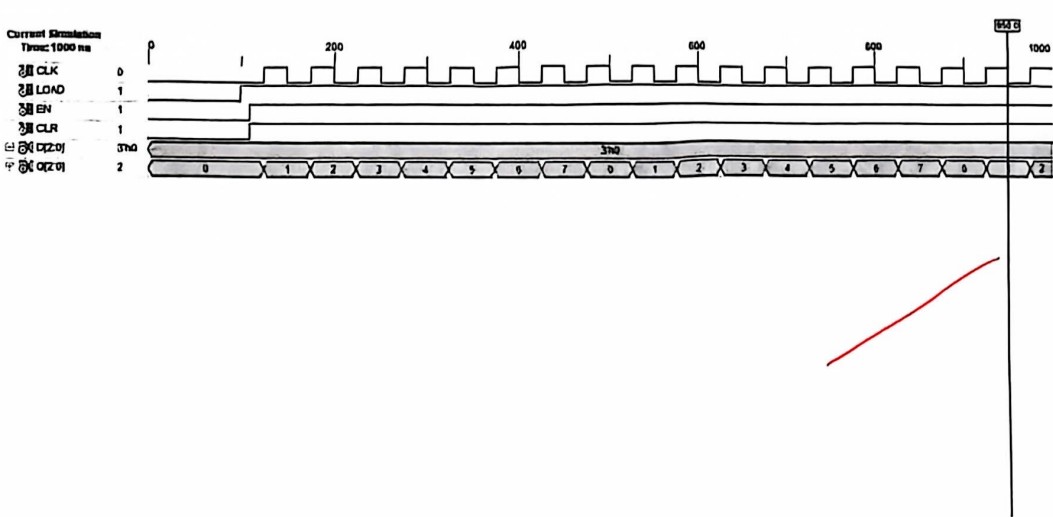
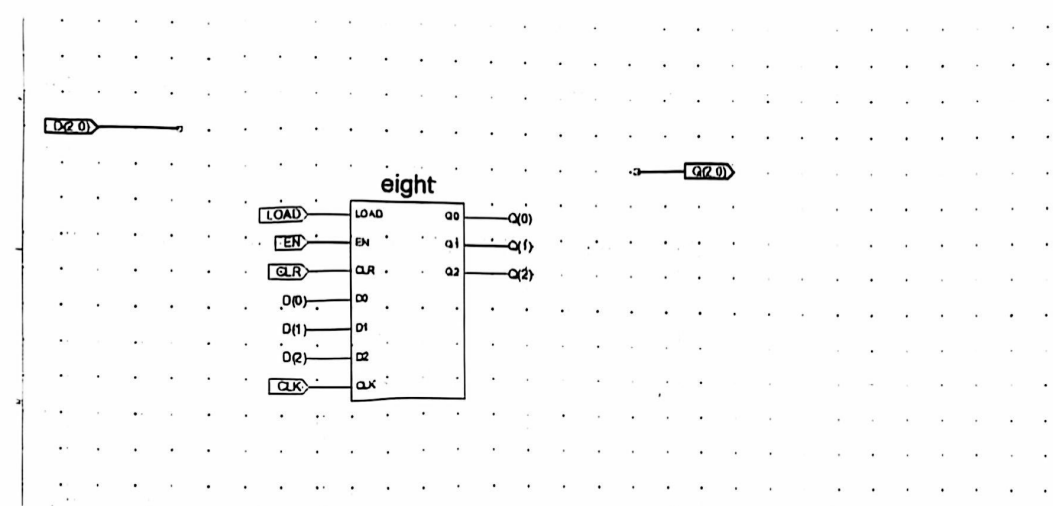
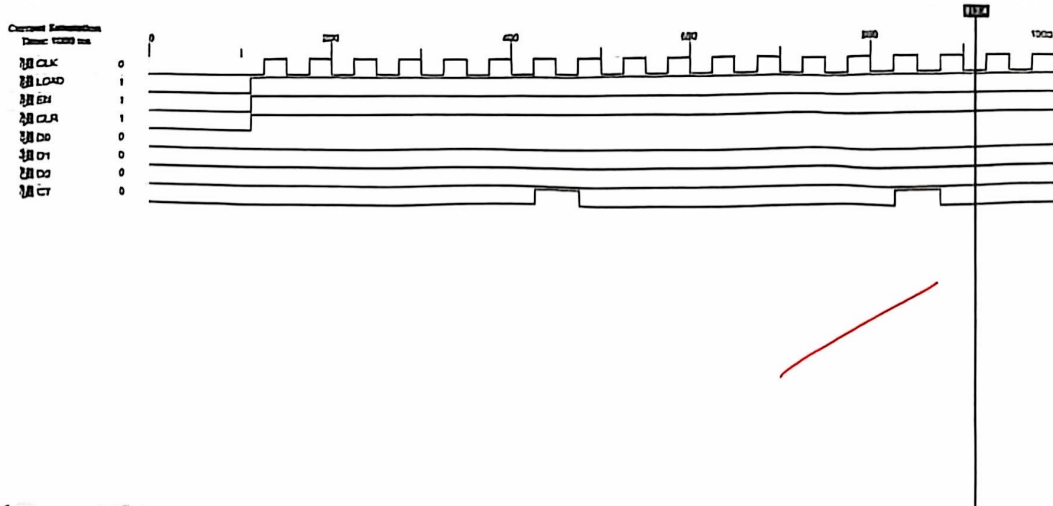
结论：

此6进制计数器与之前实验3中设计的利用8进制计数器改装的最大不同在于此计数器考虑了冗余状态，使计数器能够满足自启动的条件，而之前的那个6进制计数器则无法自启动脱离冗余状态的无效循环，需要在启动通电压人为给予一个LOAD端的置0脉冲或者CLR端的清零信号。

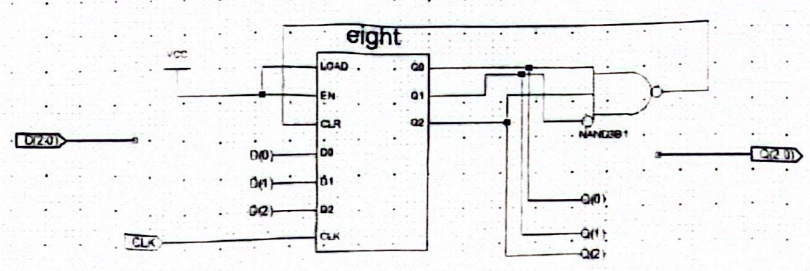
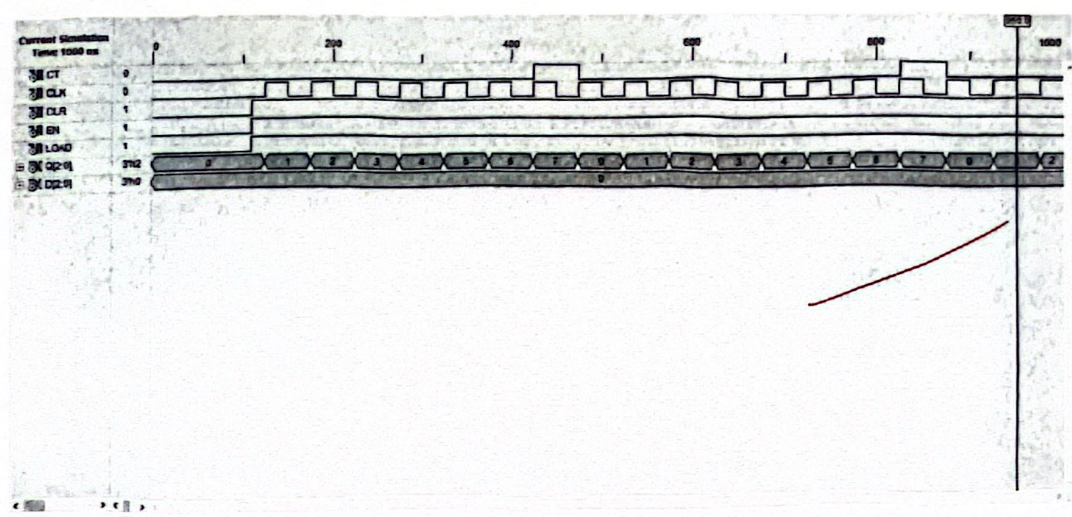
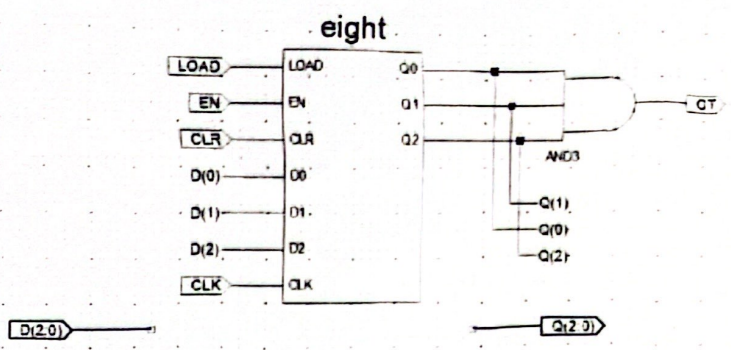
图的名字呢?

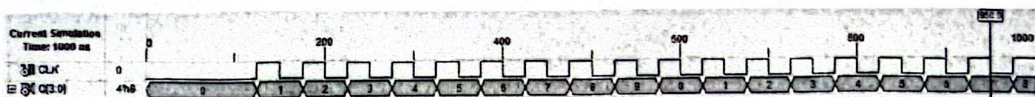
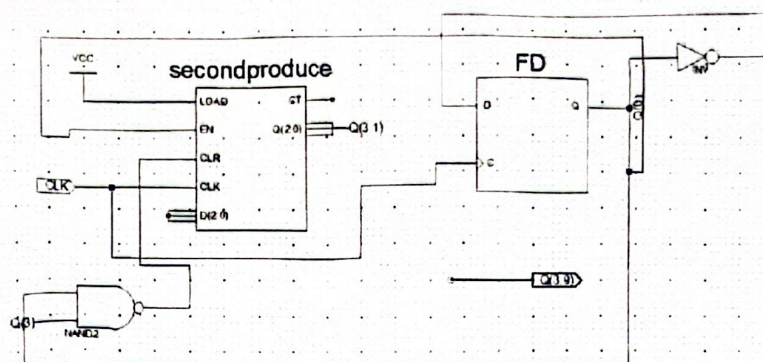


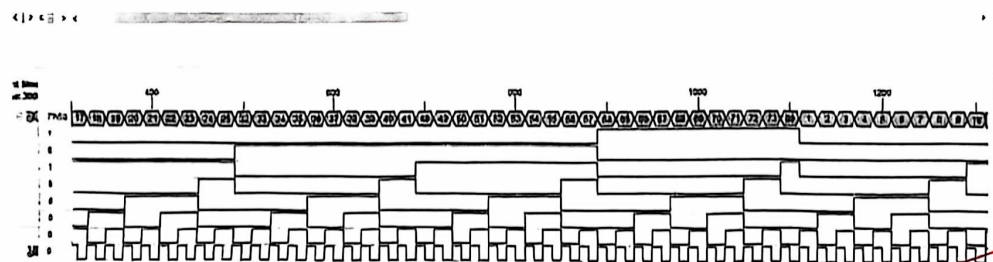
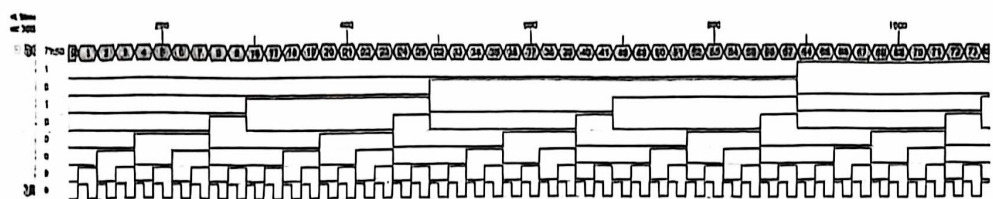
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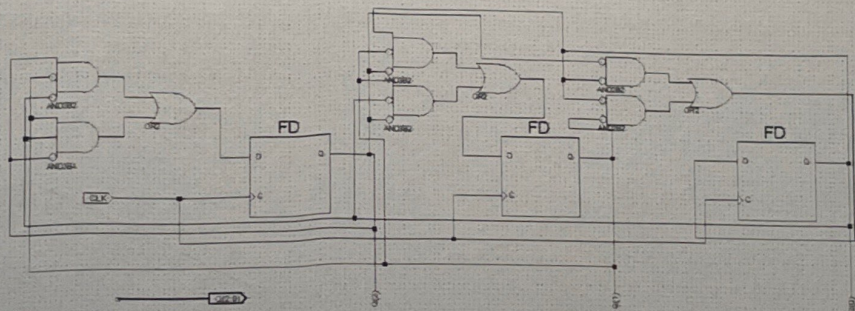
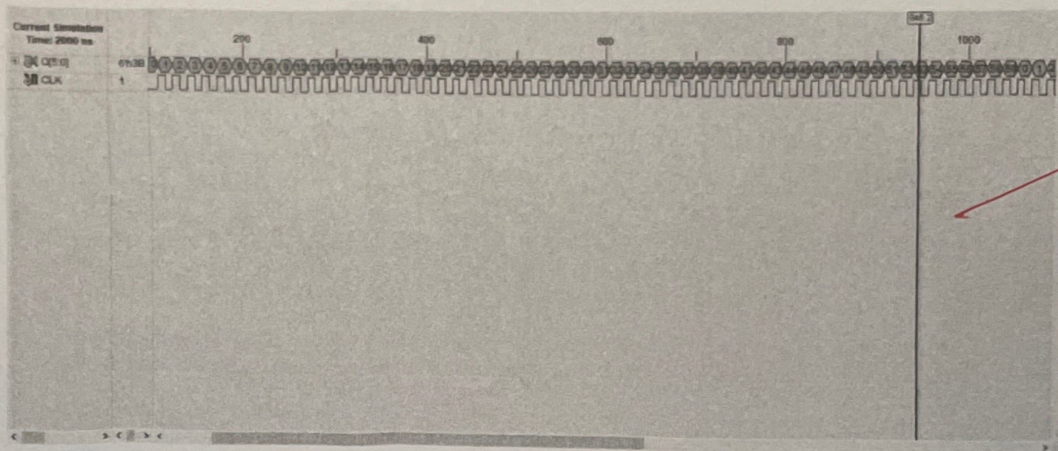
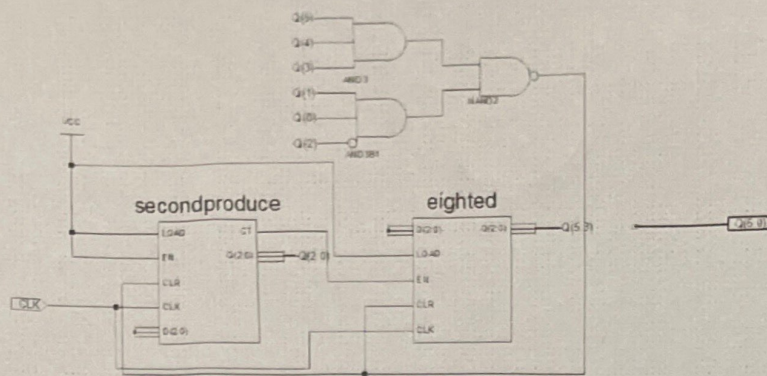


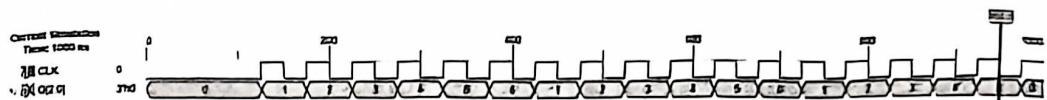
表达式
 $S = A \oplus$
 $C_0 = (A +$
 $= A$











12.2

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